

Virgo General Tokenization Notebook Summary

Notebook

virgo-general-tokenize(1).ipynb

Phase

Instruction Dataset Tokenization

Purpose

Converts the Virgo General instruction dataset into tokenized binary files for efficient language model training.

Objective

Load the formatted chat dataset, tokenize conversations using the Virgo tokenizer, create train/validation splits, and export binary datasets.

Input Resources

Virgo tokenizer (virgo_tokenizer.json) and Virgo General CSV dataset (virgo_chat_dataset.csv).

Conversation Format

Each sample is formatted as: , User:, prompt, Virgo:, response, .

Tokenizer

Custom Virgo BPE tokenizer with a vocabulary size of 45,000.

Pipeline

Load CSV → Format conversations → Shuffle dataset → Tokenize → Convert to uint16 → Split (99.5% train / 0.5% validation) → Save .bin files → Create ZIP package → Generate token statistics.

Output Files

chat_train.bin, chat_val.bin, virgo_chat_dataset.zip.

Dataset Storage

Binary token IDs stored as uint16 for compact storage and fast loading.

Engineering Highlights

Deterministic shuffling (random_state=42), tqdm progress bars, binary dataset generation, dataset packaging, token length analysis, memory-efficient storage.

Summary

This notebook prepares the Virgo General instruction dataset for training by transforming formatted conversations into compact binary token files compatible with the Virgo training pipeline.